

Yannis FERHAOUI

Software Engineer – 3D – Computer Vision

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Technical Skills

Core & 3D Languages C++, C, Python, GLSL, Qt/QML, Java

Web & Fullstack JavaScript, React.js, PHP/Symfony, HTML/CSS, PostgreSQL, MySQL

Specialized Libraries OpenGL, OpenCV, SDL2, VTK/ITK, PyTorch, TensorFlow, NumPy

Engineering & DevOps Git, GitLab CI/CD, Docker, CMake, Linux, Windows, MacOS, GoogleTest

Languages Français (native), Anglais (professional)

Education

Master's in Computer Science – Image, Development and 3D Technologies

2023 – 2025

Claude Bernard University Lyon 1, Villeurbanne

M2 : GPU programming, Virtual Reality, Image and signal processing, Computational geometry.

M1 : Software engineering & project management, Artificial Intelligence, Computational complexity, Design Patterns.

Bachelor's in Computer Science

2020 – 2023

Claude Bernard University Lyon 1, Villeurbanne

Operating systems, IP/TCP networks, SQL databases, Computer architecture, Web development, Advanced algorithms.

Professional Experience

Institut Pascal – Université Clermont Auvergne

Feb. 2025 – Jul. 2025

R&D Internship : Generative AI, Deep Learning & Software Integration

Le Puy-en-Velay

- **Design and integration** of an LLM + RAG architecture into the open-source medical software **3D Slicer** for contextual user assistance.
- **Development** of a complete extension for 3D Slicer : Qt interface, conversational agents, FAISS vector search, LangChain orchestration, and retrieval-augmented generation pipeline.

AMS Association Mantes Solidarité

May 2023 – Jun. 2023

Fullstack PHP/Symfony Developer

Mantes-La-Ville

- **Development** of a payment form and a donation management interface for administrators.

Selected Projects

Mesh Viewer | C++, Qt, OpenGL, GLSL, CMake, GitLab CI/CD

Sep. 2025 – Present

- **Development** of a high-performance 3D mesh viewer supporting standard formats (.obj, .off) with real-time rendering.
- **Design** of an OpenGL rendering pipeline (GPU loading, GLSL shaders, textures, interactive camera) and setup of a GitLab CI/CD pipeline automating cross-platform builds, testing, and deployment via CMake.

SlicerGPT | Python, Transformers, LangChain, FAISS, Qt, Hugging Face

Mar. 2025 – Oct. 2025

- **Development** of an AI extension for **3D Slicer** integrating a local conversational assistant with real-time contextual assistance.
- **Design** of a RAG pipeline leveraging the active scene, official documentation, and technical forums through FAISS, with orchestration of local LLMs using Transformers and LangChain.

Computational Geometry | C++, Data Structures, Computational Geometry

Sep. 2024 – Nov. 2024

- **Implementation and optimization** of the Lawson algorithm for the construction and manipulation of triangulated meshes.
- **Design** of efficient data structures for point cloud processing.

Embedded Robotics & Vision | C++, Linux (EV3Dev), OpenCV, Real-Time Systems

Feb. 2024 – May 2024

- **Development** of an autonomous embedded robotic system in C++ on Linux (EV3Dev), including motor control and real-time software architecture.
- **Implementation** of a computer vision pipeline using OpenCV for object detection, multi-camera fusion, and real-time image processing.